VOYNICH HERBAL A

An Induced Compositional Surface Grammar

Rules inferred from attested Voynich word forms and manuscript positions

---

Corpus-first English reference edition

31 October 2026

Purpose: to write new, freely composed Voynich-A-style passages while following explicit, auditable surface rules.

This is not a decipherment. There is no verified translation, phonology, part-of-speech system, or sentence semantics. The rules describe observed graphic and positional organization and define a deliberately conservative, usable composition standard.

Contents

1 What This Book Is For 3
1.1 Three evidence labels . . . 3
1.2 The central principle of free composition . . . 3

2 Source Corpus, Scope, and Reproducibility 3

3 Glyphs and Word Construction 4
3.1 The 23 glyphs . . . 4
3.2 The conservative word licence . . . 4
3.3 The importance of the first `q` . . . 5
3.4 What is not justified . . . 5

4 A Formal, Productive Paragraph Grammar 5
4.1 Nonterminals . . . 5
4.2 What each nonterminal means . . . 6
4.3 The slot-selection rule is conservative, not historical . . . 6
4.4 Compositional freedom . . . 6

5 Observed Preferences that Improve Manuscript-Like Style 6
5.1 First-line beginnings . . . 6
5.2 Length and line space . . . 7
5.3 Ends and closure . . . 7
5.4 W2 and the ordinary line . . . 7

6 Lexical Relatedness Without Imaginary Meanings 7

7 The Optional Family-Continuity Construction 7
7.1 How to make a topic-like graphic passage . . . 8
7.2 What the continuity evidence actually says . . . 8

8 How to Write Freely: A Practical Protocol 8
8.1 A fully free model of a line . . . 9

9 Two Completely New Grade-A Paragraphs 9
9.1 Composition I: a `q.o.t.ch` sequence . . . 9
9.2 Composition II: an `a.i.i.n` sequence . . . 9

10 A New Grade-B Form, Properly Labelled 9

11 How to Read the Lexicon and the Checker 10

11.1 The three checker grades 10

12 Limitations, Falsifiability, and Responsible Use 10

13 A One-Page Composition Checklist 11

A Most Frequent Observed Forms (an Index, Not a Semantic Dictionary) 12

B Role-Specific Word Banks for Immediate Composition 13

C High-Support Shared Paths for Optional Continuity 14

D Documented Corpus Example (not a Composition) 15

What This Book Is For

This book starts with *attested Voynich forms*, not a simulation algorithm. It gives a compositional grammar which a human writer can use to create entirely new paragraphs with independently chosen words. The rules are inferred from the supplied line-annotated transcription of *pure Herbal A*, a defined subgroup of the Voynich manuscript. They do not purport to cover every section of the manuscript.

There are two different senses of "grammar." In ordinary historical linguistics a grammar may state which expressions convey meanings. Those functions are not recoverable here. This book instead states an *auditable surface grammar*: which glyph sequences occur, how complete word forms are related, what word types occur in different positions, and how lines and paragraphs are arranged. It licenses new *texts*, not proven meaningful utterances.

Three evidence labels

OBSERVED	Direct counts, attested glyph sequences, attested word forms, manuscript-position frequencies.
LICENSED	An explicit conservative composition convention. It allows free writing within the observed inventory; it is not claimed to be a historical categorical prohibition.
EXPLORATORY	A recurrent pattern or a new analogical form supported by distributions, not a proven rule of the original script.

The central principle of free composition

Choose any topic for yourself; choose any number of paragraphs and any arrangement of the permitted words within the described slots. The slots are *positional*, not semantic. You are not required to copy or paraphrase an existing Voynich line. Word-family carryover, repetition, and morphographic alternation are available as *options*, not obligatory sentence meanings. Where corpus evidence is probabilistic, this manual does not convert it into a universal law.

Source Corpus, Scope, and Reproducibility

The source is the project-supplied, position-annotated EVA transcription, restricted to the current 85-page pure Herbal A set (Currier A). The file is bundled so that the lexical tables can be recalculated independently. The tables in this edition were made *directly from observed tokens*, without importing any VSL simulation model, candidate ranking, or generated outputs.

Observable unit	Count
Surfaces (pages)	**85**
Physical bifolia	**22**
Paragraphs	**165**
Lines	**1,090**
Word tokens	**6,836**
Attested distinct surface words	**1,971**
Observed glyph units	**23**
Observed word-internal bigrams	**229**
Observed word-internal trigrams	**881**
Shared 4–6-glyph routes (operational definition)	**715**

The 6,836-token basis is the *position-annotated, clean, 23-glyph* corpus and must not be silently exchanged with a separately normalized 6,853-token/28-glyph version. For slot statistics we use the 164 paragraphs containing at least three lines (6,816 word tokens); a two-line paragraph with 20 tokens is in the lexicon but excluded from the canonical three-stage positional analysis. The underlying EVA transcription embodies scribal and editorial choices. Counts of zero mean *not observed here*, never a proven impossibility.

Glyphs and Word Construction

The 23 glyphs

A word is displayed as an EVA string. The six composite units `cfh`, `ckh`, `cph`, `cth`, `ch`, and `sh` each count as *one glyph* for this study. The other units are single letters of the transliteration. The complete working inventory is:

```
q cfh ch cph cth p sh ckh f k o t
a d e s i l y r m n g
```

Illustrations:

Written form	Segmented glyphs	Length
`chol`	`ch-o-l`	3
`daiin`	`d-a-i-i-n`	5
`qotchy`	`q-o-t-ch-y`	5
`qotchody`	`q-o-t-ch-o-d-y`	7

The conservative word licence

Grade A: every emitted word must be one of the 1,971 *attested* surface types. This is the safest and easiest way to write freely without accidentally inventing impossible-looking local glyph joins. The annotated lexicon makes all types searchable by frequency, glyph length, and manuscript position.

Grade B: a non-attested form may be licensed by *one* local analogical edit of an attested word that is itself observed in the intended positional slot. The four supported substitution axes are `ch/sh`, `k/t`, `l/r`, and `o/a`. Single `i`-run expansion or contraction is also provisionally admitted. The resulting form must still use only the 23 units, have length 1–10 glyphs, and contain only *observed* word-internal bigrams *and* trigrams. These safeguards create a

productive analogical sublanguage; they do not prove that the new form was historically legal.

Grade C: experimental combinations using attested local n-grams without a one-edit parent. They are not part of the conservative grammar and must not be presented as attested vocabulary.

The importance of the first `q`

In the analyzed sample, 603 word tokens contain `q`; 600 begin with it, and 597 contain the immediate pair `q-o`. A writer looking for a typical form should prefer initial `qo-` to arbitrary placement of `q`. These are observed proportions, *not* an absolute `qo` command: all recorded exceptions remain part of the evidence.

What is not justified

A glyph has *not* thereby been shown to encode a sound, an inflection, a tense, a vowel, a digit, or an operation by a historical machine. A legal word-form does not acquire a known translation because it fits this chapter.

A Formal, Productive Paragraph Grammar

This is the *composition core*: a deliberately conservative, learnable subset. It is narrower than the manuscript (for example, exceptional one-word lines and two-line paragraphs exist) but not a memorized collection of sentences. A passage conforming to this grammar may use arbitrary choices from its role-indexed vocabulary.

Nonterminals

```
PARAGRAPH   := FIRST_LINE NEWLINE INNER_LINE
               (NEWLINE INNER_LINE)* NEWLINE FINAL_LINE
FIRST_LINE  := PHW F1_BODY* F1_LF
INNER_LINE  := HEAD W2 BODY* LF
FINAL_LINE  := PF_HEAD PF_BODY* PF_LF
DOCUMENT    := PARAGRAPH (BLANKLINE PARAGRAPH)*
```

Words are separated by spaces. In the print examples, a period may be used as an alternative EVA word separator. A newline terminates a physical line; a blank line terminates the paragraph. The asterisk in the schematic grammar means *any number*, including zero. The core requires at least three lines per paragraph; it does *not* prescribe a maximum number of inner lines. The number of BODY words is freely chosen.

What each nonterminal means

Slot	Corpus-based meaning
PHW	First word of the paragraph's first line; heavily biased toward gallows starts.
F1_BODY	Any interior word of the first line (including its W2).
F1_LF	The first line's final word.
HEAD	The first word of a noninitial, nonfinal line.
W2	Its second word, empirically distinguishable as Post-Head.
BODY	Interior words following W2 on a nonfinal line.
LF	The line's final word.
PF_HEAD	First word of the paragraph's last line.
PF_BODY	Interior words on that last line.
PF_LF	Final word of the entire paragraph.

Unlike an abstract statistical generator, this grammar does not require a hidden SETUP instruction, a latent probability mode, or a prescribed candidate list. Its terminals are plain observed words. The writer can independently select *any* word that appears in the chosen slot's corpus inventory. With this simple rule, whole sentences and paragraphs can be newly written; the particular sequences do not have to be manuscript quotations.

The slot-selection rule is conservative, not historical

For Grade A, require evidence that a word has occurred at least once in *that exact slot*. This produces an easily checked writing standard. It is not the claim that a word never observed at W2 could never have been written at W2 in the 15th century. A supplementary checker can relax the slot rule and report deviations as advisories rather than hard failures.

Compositional freedom

There is no imposed ordering of nouns and verbs, no subject–object template, no finite state that forces particular lexical items, and no requirement that daiin or any other frequent word appear in every paragraph. A writer may choose a line about an arbitrary private subject by associating a label with a word family *for their own composition*; such labels are not translations of the Voynich manuscript.

Observed Preferences that Improve Manuscript-Like Style

These preferences can guide a writer but are not additional pass/fail rules. Two grade-A compositions can both be formally licensed while differing greatly in their closeness to corpus tendencies.

First-line beginnings

Within the 164 paragraphs with at least three lines, 136 of 164 PHW tokens begin with a gallows glyph p/f/k/t: **82.9%**. Ordinary inner HEAD positions show only about **5.4%** such starts, and final-line PF-HEAD positions about **3.2%**. Thus it is advisable, not mandatory, to start a new paragraph with pchor, tchor, tshol, or another attested PHW form.

Length and line space

The observed average words per physical line is approximately 6.27, with a median of 6; the 25th to 75th percentiles are 5–8, and the full observed range is 1–13. A practical free-writing default is five to eight words per line. It is not a ban on longer or shorter lines. Space constraints arise from handwriting and illustration geometry; they cannot be transformed into a precise mechanical character cap using this text corpus alone.

Ends and closure

The most common word-final glyphs are `y` (2,202 observed word tokens), `n` (1,213), `l` (1,147), and `r` (1,112). Their distribution changes across line positions. In the core positional sample the mean glyph length is about 4.02 for ordinary LF and 4.63 for PF-LF. A final-line end can therefore sound visually more emphatic through attested terminal choices and length. Nothing establishes that it is a tense or predicate ending.

W2 and the ordinary line

Noninitial middle lines are advantageously written as `HEAD W2 BODY ... LF`. W2 is a *position*, not a fixed lexeme. Frequently observed W2 forms include `chol`, `chor`, `daiin`, `shol`, `chy`, and `cthy`. We do not impose a separate obligatory lexical class at the second *line*; evidence for a unique L2-W2 board is weaker than evidence for common W2/Post-Head behavior across lines.

Lexical Relatedness Without Imaginary Meanings

A writer can make new, coherent-looking local series by alternating members of attested graphic paradigms. For example:

Pair of attested words	Observed token counts	Graphical contrast
`chol / chor`	206 / 144	final `l/r`
`chol / shol`	206 / 95	initial `ch/sh`
`qokchy / qotchy`	37 / 44	interior `k/t`
`daiin / dain`	369 / 77	one `i` fewer
`dol / dor`	40 / 33	final `l/r`
`chor / char`	144 / 27	interior `o/a`

The corpus contains 136 distinct `l>r` attested pairs, 162 `ch>sh` pairs, 179 `k>t` pairs, and 153 `o>a` pairs under the recorded one-glyph substitution test. These are *pairs of existing complete forms*; the figures are not counts of conjugations or semantic agreement.

A form such as `chol` differs from `chor` in one terminal glyph. No causal direction of derivation follows: the change might encode a function, reflect a generative branch, or simply be a consequence of dense local letter constraints. This is why the dictionary uses *surface forms* instead of invented noun and verb lemmas.

The Optional Family-Continuity Construction

How to make a topic-like graphic passage

A *route* is a contiguous sequence of four to six glyphs present inside multiple complete words. For this edition we catalog a route only if it occurs within at least two distinct complete forms, on at least two manuscript surfaces, and in at least three tokens. This editorial definition recovers **715** routes from the observed sample.

The composition technique is voluntary:

1. Select one registered route in the route index, for example `q.o.t.ch` or `a.i.i.n`.
2. On line 2, choose one or more attested carrier words containing that route, from the corresponding permitted slots.
3. On line 3, choose one or more *different* carrier words containing the same route.
4. On later lines you may maintain the route, change routes, interleave others, or abandon the repetition entirely.

This makes new texts with visible family continuity and does not prescribe an invented meaning for the family. A complete surface word may contain several registered routes, and a word with fewer than four glyphs (such as `chol`) need not belong to any 4–6-glyph route, despite its high frequency and obvious graphical relationships.

What the continuity evidence actually says

In the 142 four-or-more-line paragraphs, a simple independent re-tabulation of the distinct four-glyph routes shared by line 2 and line 3 gives **161** observed intersection entries. A 5,000-shuffle comparison that preserves the paragraph and its first and last lines but re-distributes the interior words among interior line positions gives mean **142.1**; 177 of 5,000 randomizations reach or exceed 161 (one-sided plus-one $p \approx 0.0356$). This is *exploratory*, not evidence of semantic coreference. Alternative definitions of eligible shared paths have yielded other counts, so the numerical definitions must be kept separate.

Therefore a composition *without* a shared L2–L3 four-glyph route is not ungrammatical. Likewise, no evidence requires a route to make its first-ever appearance precisely on line 2.

How to Write Freely: A Practical Protocol

1. **Choose the scale.** Decide how many paragraphs and how many lines you want. Use at least three lines per paragraph for the conservative classroom grammar; the number of inner lines and BODY tokens is open.
2. **Choose a first-line entrance.** Consult the lexicon's `PHW` column. A gallows-initial form is typical, but non-gallows forms occur in the corpus too.
3. **Fill the first line freely.** Pick any attested `F1_BODY` forms and end with an attested `F1_LF` form. You need not reuse any known manuscript phrase.
4. **Compose line 2.** Pick a `HEAD`, a `W2`, any number of `BODY` forms, and an `LF`. Optionally elect a four- to six-glyph family to recur later.
5. **Compose additional ordinary lines.** Keep the slot shape, but vary words. Repeat/contrast forms if stylistically useful; optional family continuity may be present or absent.
6. **Close the paragraph.** Choose a `PF_HEAD`, optional `PF_BODY` words, and a `PF_LF`. Terminate the paragraph by a blank line.

7. **Optional creative word-formation.** Switch to Grade B only if you deliberately want new surface forms; apply exactly one supported graphic edit and check the word-internal bigram and trigram lists.
8. **Check before sharing.** Run the accompanying `check_composition.py`. Record whether your work is Grade A, B, or C and keep any private semantic glosses explicitly separate.

A fully free model of a line

The key to freedom is the unbounded `BODY*`. For example, `HEAD W2 LF` and `HEAD W2 BODY BODY BODY LF` are both permitted intermediate-line structures. No list of complete sentence templates is stored in the grammar. You select every word yourself from the lexicon, and you may independently choose its frequency level, length, and graphical family.

Two Completely New Grade-A Paragraphs

The following paragraphs were *newly composed for this book*, using only actual corpus words in the same observed positional roles. They are **not** quoted from the manuscript and have **no validated translation**. All 49 tokens pass the accompanying strict Grade-A checker. Their line lengths, lexical makeup, and optional family repetitions are examples, not compulsory templates.

Composition I: a `q.o.t.ch` sequence

```
pchor shol daiin chor chody shor
qotchy chol qotchor qotchol qotchody
qotchor qotshy daiin qotchol qotchody
qotchol chol dain dy daiin
```

Line 1 starts with the attested paragraph head `pchor`. On line 2, the head `qotchy` and the words `qotchor`, `qotchol`, `qotchody` share `q.o.t.ch`. That four-glyph path continues on line 3 through different carrier words. Line 4 selects `qotchol` as its final-line head and `daiin` as its paragraph-final word. These are evidence-based *positions*, not sentence meanings.

Composition II: an `a.i.i.n` sequence

```
tshol shol shor chor chody dy
shodaiin daiin dchor chol shodaiin chaiin
daiin shodaiin chol chody dain daiin
qokchy chor daiin shol cthy
sho chol dy chor daiin
```

Here the second and third lines share several four-glyph routes, including `a.i.i.n`. The fourth line introduces different forms without erasing the paragraph-level surface pattern. The final line ends with an attested paragraph-final `daiin`. The reader is free to create an unrelated composition that still has the same grammatical status.

A New Grade-B Form, Properly Labelled

Grade B permits limited, explicitly analogical novel words. For example, `doin` is *not attested in this 1,971-form pure-A sample*. It is one substitution `a>o` away from attested `dain`; the

resulting glyph sequence `d-o-i-n` uses recorded word-internal bigrams/trigrams, and the parent occurs in the relevant BODY slot. In the second example, `doin` may replace the interior `dain` on line 3. The supplied `grade_B_one_edit_original.eva` shows this single substitution and passes Grade B but *fails Grade A*.

This example demonstrates productive *graphic* composition only. Nothing is known about the meanings of `dain` and `doin`; the new word must not be presented as a discovered Voynich inflection or a word found in the manuscript.

How to Read the Lexicon and the Checker

The companion concordance contains all 1,971 attested surface types, indexed by empirical frequency. Each row gives the EVA form, segmented glyphs, glyph length, observed count, number of surfaces and bifolia, a source locus, and the positional evidence. The raw CSV includes ten separate positional-count columns; the printed concordance compacts the columns to the set of observed role labels.

File or action	Purpose
`attested_lexicon.csv`	All observed words with exact slot counts and reproducible loci.
`shared_routes.csv`	All 715 registered four- to six-glyph route families and representative carriers.
`word_internal_bigrams.csv`	The 229 observed adjacency types, with counts.
`word_internal_trigrams.csv`	The 881 observed three-glyph paths, with counts.
`attested_single_substitutions.csv`	Empirically co-attested analogical form pairs for four substitution axes.
`check_composition.py`	Check a human-authored text; no automatic text-generation operation.

The three checker grades

```
python tools/check_composition.py examples/grade_A_original.eva --level A
python tools/check_composition.py examples/grade_B_one_edit_original.eva --level B
python tools/check_composition.py examples/grade_A_original.eva --level A --json
python tools/check_composition.py --list-role PHW
```

For Grade A, a word must be attested *and* attested in the corresponding role. For Grade B, the same is required of the parent of any newly created word, in addition to one analogical edit and observed local glyph connections. Grade C is a flagged research extension. A checker **PASS** is never equivalent to “Voynich meaning confirmed” or “authentic manuscript sentence”.

Limitations, Falsifiability, and Responsible Use

1. **No decipherment claim.** The names `PHW`, `W2`, `LF`, and `PF_LF` describe positions, not nouns, verbs, cases, discourse particles, or tenses.
2. **No fixed word semantics.** The lexicon contains no English translations because none have been established. A private translation for a creative story remains a convention invented by the writer.

3. **Finite observation is not universal prohibition.** An unattested trigram, role, or word could appear in a lost page or a different manuscript section. Grade A is a *conservative writing standard*, not a comprehensive historical grammar.
4. **No claim of universal paragraph staging.** The first line has strong positional anomalies; "line 2 activates a topic" is a useful proposed interpretation, not a syntactic fact. Family carryover is weak and nonmandatory.
5. **No arbitrary n-gram language masquerading as historical proof.** Grade B demands a local attested analogue; Grade C is deliberately flagged. This prevents fictitious words from being silently promoted to evidence.
6. **One manuscript register.** Pure Herbal A is not identical to Herbal B, the biological/balneological section, or the zodiac and starred texts. Corpus limits must be preserved.
7. **Drawing geometry is absent.** Positions are obtained from the annotated transcription; botanical boundaries, page width, original writing order, and all scribal readings require independent image checks.

The rules are falsifiable: additional transcriptions could change the 23-glyph alphabet or the observed positional inventories, and new blinded comparisons could challenge the usefulness of the family-continuity construction. The writing guide should then be updated rather than retroactively reinterpreted as a proven ancient syntax.

A One-Page Composition Checklist

	Before calling a new passage Grade A
☐	Every word is present in `attested_lexicon.csv`.
☐	Every word occupies a role in which that exact form was observed.
☐	Every paragraph has a first line, at least one inner line, and a final line.
☐	First line: `PHW ... F1_LF`.
☐	Ordinary line: `HEAD W2 ... LF`.
☐	Final line: `PF_HEAD ... PF_LF`.
☐	Newlines mark lines; blank lines separate paragraphs.
☐	No form or word order is described as a verified meaning, tense, or part of speech.

Optional stylistic goals: gallows-first paragraph head; about 5–8 words per line; varied but related surface forms; shared four-glyph route on lines 2–3; measured terminal choices. These are not necessary for a formal Grade-A pass.

Most Frequent Observed Forms (an Index, Not a Semantic Dictionary)

The entries below are rank-ordered by observed token count in the 85-page corpus. The exact position counts are supplied in the complete companion concordance and CSV.

Rank	Attested form	Glyph units	Count	First locus
1	daiin	d.a.i.i.n	369	f1v.8
2	chol	ch.o.l	206	f1v.6
3	chor	ch.o.r	144	f2r.3
4	s	s	108	f2r.2
5	dy	d.y	105	f2r.4
6	chy	ch.y	97	f1v.6
7	shol	sh.o.l	95	f1v.5
8	sho	sh.o	90	f2r.9
9	cthy	cth.y	86	f1v.3
10	dain	d.a.i.n	77	f2r.6
11	y	y	69	f2r.3
12	dar	d.a.r	60	f1v.5
13	shor	sh.o.r	60	f1v.9
14	chey	ch.e.y	58	f1v.5
15	shy	sh.y	56	f1v.3
16	or	o.r	50	f1v.2
17	ol	o.l	49	f1v.1
18	cho	ch.o	47	f3r.4
19	o	o	47	f1v.1
20	qotchy	q.o.t.ch.y	44	f4r.5
21	cthol	cth.o.l	42	f1v.8
22	dal	d.a.l	40	f1v.3
23	dol	d.o.l	40	f1v.4
24	cthor	cth.o.r	39	f2r.12
25	chaiin	ch.a.i.i.n	37	f2r.3
26	oty	o.t.y	37	f6v.5
27	qokchy	q.o.k.ch.y	37	f3v.13
28	aiin	a.i.i.n	36	f3r.20
29	d	d	33	f2r.4
30	dor	d.o.r	33	f2v.5
31	r	r	33	f3v.4
32	oky	o.k.y	31	f3r.8
33	shey	sh.e.y	30	f2v.5
34	dam	d.a.m	29	f1v.4
35	kchy	k.ch.y	29	f2v.2
36	otchy	o.t.ch.y	28	f2v.3
37	char	ch.a.r	27	f1v.1
38	okaiin	o.k.a.i.i.n	27	f4r.4
39	otol	o.t.o.l	27	f5r.2
40	qoty	q.o.t.y	27	f2v.2
41	cheor	ch.e.o.r	26	f2v.4
42	otaiin	o.t.a.i.i.n	26	f2v.1
43	dair	d.a.i.r	25	f1v.4
44	chody	ch.o.d.y	24	f1v.2
45	kol	k.o.l	24	f1v.8

Rank	Attested form	Glyph units	Count	First locus
46	otchol	o.t.ch.o.l	24	f3r.7
47	dchy	d.ch.y	23	f2r.6
48	dchor	d.ch.o.r	22	f1v.9
49	ckhy	ckh.y	21	f1v.3
50	qokol	q.o.k.o.l	21	f3r.5
51	qoky	q.o.k.y	21	f7v.2
52	tchy	t.ch.y	21	f4r.9
53	choty	ch.o.t.y	20	f7r.8
54	okol	o.k.o.l	20	f1v.9
55	chodaiin	ch.o.d.a.i.i.n	19	f2v.4
56	kchol	k.ch.o.l	19	f8v.7
57	l	l	19	f1v.1
58	odaiin	o.d.a.i.i.n	19	f3r.20
59	chcthy	ch.cth.y	18	f2r.13
60	cheol	ch.e.o.l	18	f1v.7
61	choky	ch.o.k.y	18	f1v.6
62	qotol	q.o.t.o.l	18	f4v.11
63	sheey	sh.e.e.y	18	f2r.9
64	ar	a.r	17	f6v.15
65	cthey	cth.e.y	17	f1v.7
66	qotor	q.o.t.o.r	17	f5v.4
67	cheey	ch.e.e.y	16	f7v.3
68	ctho	cth.o	16	f1v.6
69	kchor	k.ch.o.r	16	f2v.5
70	ky	k.y	16	f1v.8
71	qo	q.o	16	f1v.7
72	saiin	s.a.i.i.n	16	f2r.6
73	sy	s.y	16	f8r.3
74	ykchy	y.k.ch.y	16	f3v.8
75	dchol	d.ch.o.l	15	f2r.6
76	kaiin	k.a.i.i.n	15	f8r.19
77	okal	o.k.a.l	15	f1v.6
78	qokaiin	q.o.k.a.i.i.n	15	f5r.2
79	chear	ch.e.a.r	14	f3v.4
80	otchor	o.t.ch.o.r	14	f3r.19

Role-Specific Word Banks for Immediate Composition

Each bank lists the most frequently observed forms in its own slot; you may use any other lexicon entry with a positive count in that same role.

PHW (164 tokens, 138 types)

tchor pchor tshol shol pchocthy kooiin kchor pcheol pydaiin pchar paiin tchol pcho tshor

F1_BODY (848 tokens, 478 types)

daiin shol shor chor shy sho shey dy chol chey s o chy oty

F1_LF (164 tokens, 130 types)

daiin s dy shy dain am shey shor chdy opchordy sy dal kchy oty

HEAD (760 tokens, 425 types)

daiin sho or qokchy chol qotchy dchor shor otchol qotor shol dor ychor qo

W2 (749 tokens, 345 types)

chol chor daiin sho shol chy cho cthy ol shor shy char chey y

BODY (2669 tokens, 811 types)

daiin chol chy chor s shol dain dy cthy sho chey y dar dol

LF (759 tokens, 406 types)

daiin dy cthy s dal dar dam dain aiin chaiin d dan y oty

PF_HEAD (158 tokens, 116 types)

daiin sho chor dain ycheor or ysho dsho qokol qotchy qotchor sor dol otol

PF_BODY (381 tokens, 196 types)

chol daiin chor chy y s cheor dy chey kchy dain cheey cthy o

PF_LF (164 tokens, 128 types)

daiin dy cthy chor dan chom chckhy okain cthor shy chkaiin otal dar chol

High-Support Shared Paths for Optional Continuity

A route is a contiguous sequence of *glyphs*, so a dot in the following index is a glyph boundary, not a space between complete words. These are form-level recurrence inventories, not translations or topics.

Glyph path	Tokens	Types	Sample carrier
a.i.i.n	891	161	daiin
d.a.i.i	527	68	daiin
d.a.i.i.n	490	47	daiin
o.d.a.i	109	46	chodaiin
q.o.t.ch	101	19	qotchy
k.a.i.i	98	22	okaiin
d.a.i.n	97	19	dain
o.d.a.i.i	94	32	chodaiin
o.t.ch.o	93	23	otchol
o.t.ch.y	92	11	qotchy
k.a.i.i.n	88	14	okaiin
o.d.a.i.i.n	88	26	chodaiin
q.o.k.ch	83	19	qokchy

Glyph path	Tokens	Types	Sample carrier
t.a.i.i	77	14	otaiin
o.i.i.n	76	28	choiin
t.a.i.i.n	75	13	otaiin
o.k.ch.o	74	25	qokchol
o.k.ch.y	72	9	qokchy
o.t.o.l	67	15	otol
ch.a.i.i	66	23	chaiin
ch.a.i.i.n	66	23	chaiin
o.k.a.i	65	13	okaiin
o.k.a.i.i	57	9	okaiin
ch.o.d.y	56	22	chody
t.ch.o.l	55	9	otchol
k.ch.o.l	53	12	kchol
o.k.a.i.i.n	53	6	okaiin
q.o.t.o	53	15	qotol
t.ch.o.r	53	10	tchor
k.e.e.y	52	13	qokeey
o.t.a.i	50	8	otaiin
ch.o.d.a	49	23	chodaiin
o.k.o.l	48	9	qokol
k.ch.o.r	47	11	kchor
o.t.a.i.i	47	6	otaiin
q.o.k.o	47	14	qokol
o.t.a.i.i.n	45	5	otaiin
q.o.t.ch.y	45	2	qotchy
ch.e.e.y	44	20	cheey
ch.e.o.r	44	12	cheor
o.k.e.e	43	23	qokeey
o.l.d.y	42	28	choldy
ch.e.o.l	41	16	cheol
t.ch.e.y	39	10	otchey
o.t.ch.o.l	38	5	otchol

Documented Corpus Example (not a Composition)

For comparison, the first paragraph of `f1v` in the bundled annotated dataset begins with these words, preserving line boundaries but replacing EVA separators with spaces:

```
kchsy chydaiin ol o l tchey char cfhar am
yteey char or ochy dcho lkody okodar chody
da ckhy ckhockhy shy dksheey cthy kotchody dal
dol chokeo dair dam sochey chokody
```

The paragraph is included as a short *observed attestation* for structural comparison. The two example paragraphs in Chapter 9 were newly arranged and are not manuscript quotations. No semantic reading is assigned to either.